

L^AT_EX File Template

A guide to its setup, formatting, and custom commands.

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This version of the documentation is accurate as of Version 3.0.0 of the template.

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If there are any issues, contact me using one of the below options:
[GitHub](#) or [LinkedIn](#).

Abstract

Thank you for choosing to use L^AT_EX File Template. If you find any bugs or want some feature not currently included, then my contact information can be found on [my GitHub page](#). For the avoidance of doubt, while my time at Durham University may be mentioned in places in this document, this is an independent project, and as such is in no way associated with the university.

This template was designed to create a structured starting point for academics (especially those studying mathematics or computer science) to create professional, well-formatted documents that follow traditional dissertation guidelines. This repository contains the actual template, documentation files (linked below), and their supporting build files, demonstrating the capabilities of the template. This version of the documentation is accurate for Version 3.0.0 of the template.

Acknowledgements

I would like to thank Dr Jamie Mason for his support and expertise in leading my discrete mathematics tutorials and seminars during my first year at Durham University, without which this project would probably not exist. His continuous feedback throughout has allowed me to hone the project from the messy original versions into a useable tool.

Thanks also to Joseph Eddon for supporting the development of this template through the creation of his own L^AT_EX template, and for the feedback and suggestions he provided on multiple versions of the template throughout last year.

Finally, thanks to James Leatherland for his support and feedback throughout the first year of this project.

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Chapter 1

Template Setup

1.1 Choosing your Compiler

1.2 The Template Files

1.2.1 The Main Files

1.2.2 The Template Setup Folder

1.2.3 The Chapters Folder

1.3 Additional Files

1.3.1 Working with GitHub

1.3.2 Inserting Figures

1.3.3 Creating a Bibliography

1.4 Packages List

To create this template, I had to select a variety of packages to be used throughout to provide additional functionality for formatting and custom commands. Below is a list of the packages installed, what they do, any additional documentation required, and credit to the original creators.

Chapter 2

Formatting Options

2.1 Document Formatting

2.1.1 The Title Page

2.1.2 Changing the Colour Scheme

2.2 Additional Sections

2.2.1 The Prelude

2.2.2 The Appendix

Chapter 3

Custom Commands

3.1 Text Formatting

3.1.1 My Custom Date Formats

3.1.2 Using the Custom Text Formats

3.2 General Commands

3.2.1 Figures and Tables

3.2.2 Academic Environments

3.2.3 Referencing Environments

3.3 Mathematics and Computer Science

3.3.1 Additional Mathematics Commands

3.3.2 Computer Science Environments

Chapter 4

The Bibliography

The template is set up with an automatically populating bibliography using BibL^AT_EX. This allows you to maintain and organise your citations all in one place with consistent formatting being created by the library and my formatting settings. At the moment, the template does not contain any citations, and to prevent errors, I have had to comment out some lines. These are line 10 on `file-name.tex` (shown below as [Algorithm 1](#))

Algorithm 1 Commented bibliography lines (`file-name.tex`).

```
\addbibresource{references.bib} % Uncomment to add a bibliography.
```

and lines 5-9 on `Chapters/appendix.tex` (shown below as [Algorithm 2](#)).

Algorithm 2 Commented bibliography lines (`Chapters/appendix.tex`).

```
\begin{group}
\section{Bibliography}
\defbibheading{bibliography}[]{}
\printbibliography
\end{group}
```

However, as soon as you want to add a citation, simply uncomment the lines, and then insert it into the bibliography as described in [Section 4.2](#).

4.1 Formatting in your Document

4.1.1 Citation Format

To ensure that the citations are standardised throughout the document, I have chosen to default to Harvard style citations for the project (or as close as is possible). As such, when citing one of the bibliography entries there are two in-text options, depending on how you are referring to it.

- If you are referencing something directly, you should use the format *author*, *author (year published)* (e.g., according to *author (year published)*). This is implemented... [documentation for the function \(maybe in the custom commands area\)](#).
- Conversely, if you are referencing something indirectly, you should use the format *(author, year published)* (e.g., according to research *(author, year published)*). This is implemented... [documentation for the function \(maybe in the custom commands area\)](#).

4.1.2 Bibliography Format

When designing this version of the template, I chose to make the bibliography as customisable as I could, whilst maintaining the simplicity of the system. As such, the bibliography has been set up to be completely unsorted. This allows the user to choose how to sort it, rather than forcing them into a set version. The drawback of this system is that the user does then have to sort the entries themselves but, on balance, I believe that this is a good tradeoff.

4.2 The Bibliography File

Entries into the bibliography are stored in `references.bib` and are then automatically pulled through into the bibliography (in the appendix). To add a new entry, follow the guide below about the entry formats and the general rules around entries. Please note that this is not a definitive list and can be adapted using the more comprehensive guide at Kime, Wemheuer, and Lehman (2026).

4.2.1 Entry Formatting

BibL^AT_EX provides a variety of entry types, for simplicity's sake, I have listed them below, alongside a description of their use, and their required fields for Harvard references as seen in the tables below. This was produced in combination between OpenAI (2026) and Kime, Wemheuer, and Lehman (2026).

Source Type	Entry Type	Required Fields	Optional Fields
Book	@book	title, author, date.	editor, edition, chapter, pages, publisher, location, isbn, doi, url, urldate.
Dictionary/Encyclopedia Entry	@inreference	author, booktitle, title, date.	editor, edition, chapter, pages, publisher, location, isbn, doi, url, urldate.
Film/Movie	@movie	title, author, date.	publisher, location, url, urldate.
Interview	@misc	author, date, howpublished.	title, organization, url, urldate.
Letter	@letter	author, date, howpublished.	title, organization, url, urldate.
Manual/Documentation	@manual	title, author, date.	version, url, urldate.
News Article	@article or @online	title, author, journaltitle, date.	organization, pages, url, urldate.
Speech	@unpublished	author, date, eventtitle.	title, venue, organization, url, urldate.

Table 1: biblatex .bib file entry formats (traditional media).

Source Type	Entry Type	Required Fields	Optional Fields
AI Chat	@online	author, title, date, note (with model), url, urldate, organization.	n/a.
Blog Post	@online	author, title, date, url, urldate.	organization.
Podcast/Audio Recording	@audio	author, title, date, url, urldate.	publisher, series, number.
Social Media Post	@online	author, date, organization, url, urldate.	n/a.
Software	@software	author, title, version, date.	organization, url, urldate.
Website	@online	author, title, date, url, urldate.	n/a.

Table 2: biblatex .bib file entry formats (online).

Source Type	Entry Type	Required Fields	Optional Fields
Legal Case	@jurisdiction	author, title, date, institution.	url, urldate.
Legislation	@legislation	title, date.	section, institution, url, urldate.
Patent	@patent	author, title, date, number.	location, organization, url, urldate.

Table 3: biblatex .bib file entry formats (legal/political).

Source Type	Entry Type	Required Fields	Optional Fields
Article (Journal)	@article	author, title, date.	journaltitle, volume, number, pages, doi, url, urldate.
Conference Paper	@inproceedings	author, title, date, eventtitle.	venue, pages, publisher, pages, doi, url, urldate.
Dataset	@dataset	author, title, date.	publisher, version, doi, url, urldate.
Document (General)	@misc	author, title, date.	howpublished, publisher, location, number, url, urldate.
Presentation/Conference Talk	@unpublished	author, title, date, eventtitle.	venue, organization, url, urldate.
Report	@report	author, title, date.	organization, number, type, publisher, location, url, urldate.
Thesis/Dissertation	@thesis	author, title, institution, date.	doi, url, urldate.

Table 4: biblatex .bib file entry formats (academic).

4.2.2 Field Formats

OpenAI (2026) and Kime, Wemheuer, and Lehman (2026) also defines the data types and formats of each of the fields. Below, I have listed a summary of the formats and their rules. An example of a complete entry can be seen in [Algorithm 3](#).

- Citations should be identified with unique keys – ideally title.description.
- All fields are surrounded in curly brackets.
- Where the author is an organisation, you may need to add an additional set of curly brackets to ensure it is kept together as one entry.
- Multiple authors should be separated with 'and's.
- Dates are formatted as YYYY-MM-DD.

Algorithm 3 Example biblatex .bib entry.

```
@manual{biblatex_manual,
  author = {Kime, Philip and Wemheuer, Moritz and Lehman, Philipp},
  title = {The biblatex Package},
  date = {2026},
  version = {3.22a},
  url = {https://tex.org.uk/macros/latex/contrib/biblatex/doc/biblatex.pdf},
  urldate = {2026-10-10}
}
```

Chapter 5

Appendix

5.1 Bibliography

Kime, Philip, Moritz Wemheuer, and Philipp Lehman (2026). *The biblatex Package*. Version 3.22a. URL: <https://tex.org.uk/macros/latex/contrib/biblatex/doc/biblatex.pdf> (visited on 10/10/2026).

OpenAI (2026). *BibLaTeX Reference Guide for Harvard-style Citations*. Model is GPT-6 Luna. OpenAI. URL: https://chatgpt.com/s/t_6aca1d00deb48191a4a0d9e4498cd5ea (visited on 10/10/2026).

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4	biblatex .bib file entry formats (academic).	6

5.3 List of Figures